

WORKSHEET 02

CUQ

- Slope of the line with inclination 60° is
 - $\frac{1}{\sqrt{3}}$
 - $\sqrt{3}$
 - 1
 - 0
- Slope of the line passing through (b, b^2) and (a, a^2) is
 - $a - b$
 - $b - a$
 - $a + b$
 - $a^2 + b^2$
- Slope of the line $ax + by + c = 0$ is
 - $m = \frac{+a}{b}$
 - $m = \frac{-b}{a}$
 - $m = \frac{-a}{b}$
 - $m = \frac{b}{a}$
- If $ax + by + c = 0$ is the general form of the line, then
 - $|a| + |b| = 0$
 - $|a| + |b| \neq 0$
 - $a + b = 0$
 - $a + b \neq 0$
- If $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are two parallel lines, then the correct statement is
 - $a_1 = a_2, b_1 = b_2, c_1 \neq c_2$
 - $a_1 \neq a_2, b_1 = b_2, c_1 = c_2$
 - $a_1 \neq a_2, b_1 \neq b_2, c_1 = c_2$
 - $a_1 = a_2, b_1 \neq b_2, c_1 = c_2$
- Slope - intercept form of the line is
 - $y = mx$
 - $y = mx + c$
 - $y = mx - c$
 - $y - y_1 = m(x - x_1)$
- Equation of the line with slope 'm' and x - intercept 'a' is
 - $y = mx + a$
 - $y = my + a$
 - $y = m(x - a)$
 - $y = m(x + a)$

JEE MAIN

LEVEL - 1

Single Correct Choice Type:

- The equation of the line which makes 45° with x-axis and passes through $(1, 0)$ is
 - $x + y - 1 = 0$
 - $x - y - 1 = 0$
 - $x + y + 1 = 0$
 - $x - y + 1 = 0$
- The equation of the line having intercepts 1 and 2 respectively is
 - $x + 2y - 2 = 0$
 - $x + y = 2$
 - $2x + y - 2 = 0$
 - $x - y = 2$
- The slope of the line joining the points $(-3, 2)$ and $(5, -4)$ is
 - $-3/4$
 - $3/4$
 - $4/3$
 - $-4/3$

4. The slope of the line $\frac{x}{a} + \frac{y}{b} = 1$ is
- 1) $\frac{-a}{b}$ 2) $\frac{-b}{a}$ 3) $\frac{a}{b}$ 4) $\frac{b}{a}$
5. The equation of the straight line through the points (3, 3) and (7, 6) is
- 1) $3x - 4y + 3 = 0$ 2) $3x + 4y - 3 = 0$ 3) $3x - 3y - 1 = 0$ 4) none
6. If the slope of a line is $\frac{-1}{\sqrt{3}}$, then its inclination is
- 1) $\frac{\pi}{3}$ 2) $\frac{5\pi}{6}$ 3) $\frac{2\pi}{3}$ 4) $\frac{3\pi}{4}$

JEE MAIN**LEVEL - 2**

7. The inclination of line passing through (2, 1) and $(0, 1 - 2\sqrt{3})$ with +ve X-axis is
- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{6}$ 4) $3\frac{\pi}{4}$
8. Equation of the straight line joining the points $[am_1^2, 2am_1]$ and $[am_2^2, 2am_2]$
- 1) $2x + (m_1 + m_2)y - 2am_1m_2 = 0$ 2) $2x - (m_1 + m_2)y + 2am_1m_2 = 0$
 3) $2x - (m_1 + m_2)y - 2am_1m_2 = 0$ 4) none
9. If the lines $2x + 3y - 10 = 0$ and $3x - ay - 25 = 0$ are perpendicular, then the value of 'a' is
- 1) 3 2) 2 3) -3 4) -2
10. If the lines $x - 3y = 10$ and $kx + 6y = 1$ are parallel, then the value of k =
- 1) 2 2) -2 3) 1 4) -1

JEE MAIN**LEVEL - 3**

11. The equation $8x + 6y - 15 = 0$ in slope- intercept form is
- 1) $y = 3x + \frac{5}{3}$ 2) $y = \frac{3}{4}x + 2$ 3) $y = \frac{5}{2}x + 5$ 4) $y = \frac{-4}{3}x + \frac{5}{2}$
12. The equation of perpendicular bisector of line passing through (1, 2) and (3, 4) is
- 1) $2x - y + 5 = 0$ 2) $x + y = 5$ 3) $3x - 2y + 5 = 0$ 4) $x + y = 4$
13. The equation of line passing through (2, -3) and parallel to line joining points (1, 2) & (-1, 5) is
- 1) $3x - 2y = 0$ 2) $2x - 3y = 0$ 3) $3x + 5y = 0$ 4) $3x + 2y = 0$

14. The vertices of $\triangle ABC$ are $A(1,1), B(-3,4), C(2,-5)$, then the equation of altitude through 'A' is

- 1) $5x+9y+4=0$ 2) $5x+9y-4=0$ 3) $5x-9y+4=0$ 4) $5x-9y-4=0$

JEE MAIN**LEVEL - 4**

15. If the slope of the line $\left(\frac{1}{a} + \frac{k}{b}\right)x + \left(\frac{1}{b} + \frac{k}{a}\right)y - (1+k) = 0$ is -1 , then the value of k is

- 1) 2 2) -1 3) 1 4) -2

16. $P(\alpha, \beta)$ lies on the line $y = 6x - 1$ and $Q(\beta, \alpha)$ lies on the line $2x - 5y = 5$. Then the equation of the line PQ is

- 1) $2x + y = 3$ 2) $3x + 2y = 5$ 3) $x + y = 6$ 4) $3x + y = 7$

17. The equation of the line passing through the points $(a, a^2), (b, b^2)$ is

- 1) $(a+b)x + y + ab = 0$ 2) $(a-b)x - y - ab = 0$
3) $(a+b)x - y - ab = 0$ 4) $(a+b)x + y + ab = 0$

18. The lines $2x+3y=6$, $2x+3y=8$ cut the X-axis at A, B respectively. A line $L = 0$ drawn through the point $(2,2)$ meets the X-axis at C in such a way that abscissa of A, B, C are in arithmetic Progression, then the equation of the line L is

- 1) $2x+3y=10$ 2) $3x+2y=10$ 3) $2x-3y=10$ 4) $3x-2y=10$

JEE MAIN**LEVEL - 5**

19. The equation of a line having the slope ' $-\tan\theta$ ' and passing through the point $(a^2 \sin\theta, a^2 \cos\theta)$ is

- 1) $x \sin\theta + y \cos\theta = a$ 2) $x \cos\theta + y \sin\theta = a$
3) $x \sin\theta + y \cos\theta = a^2$ 4) $x \cos\theta + y \sin\theta = a^2$

20. The perpendicular bisector of the line segment joining $P(1,4)$ and $Q(K,3)$ has Y intercept -4 , then a possible value of K is

- 1) -4 2) 1 3) 2 4) -2

21. The equation of straight line passing through $(3,2)$ and sum of whose intercepts 10 is

- 1) $3x+2y=12$ 2) $\frac{x}{2} + \frac{y}{3} = 21$ 3) $2x+3y=12$ 4) $\frac{x}{6} + \frac{y}{4} = 2$

JEE ADVANCED**LEVEL - 1****Multi Correct Choice Type:**

22. If a line passes through (2,3) and (2,-6), then

- 1) The line is parallel to x - axis 2) The line is parallel to y - axis
3) Slope of the line is undefined 4) Slope of the line is zero

Statement Type:

23. **Statement-I:** Equation of a line parallel to $2x + 3y + 6 = 0$ and passing through (3, -2) is $2x + 3y = 0$.

Statement-II: The equation of line parallel to $ax + by + c = 0$ and passing through (x_1, y_1) is $a(x + x_1) + b(y + y_1) = 0$.

- 1) Both Statements are True
2) Both Statements are False
3) Statement I is true, Statement II is false.
4) Statement I is false, Statement II is true.

Comprehension Type:

$\frac{x}{a} + \frac{y}{b} = 1$ is known as intercept form, it meet the x-axis at (a,0) and y-axis at (0,b)

24. The line $2x + 4y = 8$ meets the x-axis at

- 1) (4, 0) 2) (0, 2) 3) (2, 4) 4) (0, 0)

25. If a line passes through (3, 0) and (0, 4), then the equation of the line is

- 1) $\frac{x}{2} + \frac{y}{3} = 1$ 2) $\frac{x}{3} + \frac{y}{4} = 1$ 3) $\frac{x}{-2} - \frac{y}{-1} = 1$ 4) $\frac{x}{2} - \frac{y}{4} = 1$

26. The sum of intercepts of the line $x + y - 10 = 0$ is

- 1) 30 2) 40 3) 20 4) -20

Matrix Match Type:

27. **Column-I**

- a) The slope of the line joining the points (3, 4) and (4, 1) is
b) The slope of the line parallel to $2x + 3y + 4 = 0$ is
c) The slope of the line perpendicular to $x - 2y + 5 = 0$
d) If the line $2x - 3y = k$ passes through the origin, then k is

Column-II

- p) 0
q) $-2/3$
r) -3
s) -1
t) -2

Numerical Value Type:

28. The slope of the equation of the line $8x - 4y + 6 = 0$ is _____.

JEE ADVANCED**LEVEL-2&3****Multi Correct Choice Type:**

29. If $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ are two perpendicular lines having slopes m_1 & m_2 , then

- 1) $m_1m_2 = -1$ 2) $a_1a_2 + b_1b_2 = 0$ 3) $a_1b_2 + a_2b_1 \neq 0$ 4) $a_1a_2 - b_1b_2 = 0$

Comprehension Type:

* Equation of a line with slope 'm' and the y-intercept 'c' is $y = mx + c$

* Equation of a line with slope 'm' and the x-intercept 'a' is $y = m(x - a)$

30. The equation of the straight line cutting off an intercept 8 on x-axis and making an angle of 60° with the positive direction of Y-axis is

- 1) $x - \sqrt{3}y - 8 = 0$ 2) $x + \sqrt{3}y = 8$ 3) $y - \sqrt{3}x = 8$ 4) $y + \sqrt{3}x = 8$

31. The equation of the straight line which cuts off an intercept 3 from positive direction of the y-axis and is inclined at an angle 60° with the positive direction of x-axis is

- 1) $\sqrt{3}x - y - 3 = 0$ 2) $\sqrt{3}x - y + 3 = 0$ 3) $x - \sqrt{3}y - 3 = 0$ 4) $x + y\sqrt{3} = 3$

32. The equation of the line intersecting the y - axis at a distance of 2 units above the origin and making an angle of 30° with the positive direction of the x - axis is

- 1) $x - \sqrt{3}y + 2 = 0$ 2) $x - \sqrt{3}y + 2\sqrt{3} = 0$
3) $\sqrt{3}x - y + 2\sqrt{3} = 0$ 4) None of these

JEE ADVANCED**LEVEL-4&5****Matrix Match Type:**

33. **Column-I**

- a) Equation of line passing through (1, 2) and parallel to $x + y + 4 = 0$ is
b) Equation of line passing through (-1, 5) and perpendicular to line $2x - 3y + 4 = 0$ is
c) Intercept form of a line $2x + 3y = 12$ is
d) Equation of line making 45° with X- axis & passing through (3, 4) is

Column-II

- p) $3x + 2y - 7 = 0$
q) $x + y - 3 = 0$
r) $x - y + 1 = 0$
s) $\frac{x}{6} + \frac{y}{4} = 1$

SEEKING TO DESCRIBE

34. If the line $2x + 3y - 4 = 0$ is perpendicular to $3x - ay + 5 = 0$, then find the value of 'a'.
35. If the lines $kx + 3y + 5 = 0$ and $3x + 9y - 10 = 0$ are parallel, then find the value of 'k'.
36. Find the equation of the line having slope-2 and passing through $(0, -2)$.

STUDENT ADDITIONAL TASK

- The distance between the points $(\sqrt{3}+1, \sqrt{2}-1)$ and $(\sqrt{3}-1, \sqrt{2}+1)$ is
 1) $\sqrt{3}$ 2) $2\sqrt{3}$ 3) $2\sqrt{2}$ 4) $\sqrt{2}$
- A line segment of length 10 units and one end is $(2, -3)$. If the abscissa of the other end is 10, then its ordinate is
 1) 3 2) -6 3) -9 or 6 4) 3 or -9
- The triangle with the vertices $(2, 4), (2, 6)$ and $(2 + \sqrt{3}, 5)$ is
 1) equilateral triangle 2) scalene triangle
 3) isosceles triangle 4) right angled triangle
- Assertion (A):** If 'a' is rational number, then the triangle formed by the points $(2, 2), (3, 3)$ and (a, a) cannot be an equilateral triangle for any value of 'a'.
Reason (R): The co-ordinates of the vertices of an equilateral triangle are not all rational numbers.
 1) Both A and R are true, R is the correct explanation of A.
 2) Both A and R are true, But R is not a correct explanation of A.
 3) A is true, R is false
 4) A is false, R is true
- If $A(-3, -2), B(3, 2), C(-2, \sqrt{3}, 3\sqrt{3})$ are the vertices of a triangle, then show that $\triangle ABC$ is an equilateral triangle.
- If the triangle formed by $A(8, -10), B(7, -3)$ and $C(0, p)$ is a right angle triangle, right angled at B. Then find the value of 'p'.
- Find the area of the quadrilateral formed by the points $(1, 0), (0, 1), (-1, 0)$ and $(0, -1)$ taken in order.
- If $A(-1, 2), B(4, 1), C(7, 16)$ are the three vertices of a parallelogram ABCD, then find the coordinates of the fourth vertex D, and also its area.
- If the three vertices of a parallelogram are $(a+b, a-b), (2a+b, 2a-b), (a-b, a+b)$, then the fourth vertex is
 1) (b, b) 2) $(-b, b)$ 3) $(b, -b)$ 4) $(-b, -b)$
- P, Q, R are the mid points of AB, BC, CA of $\triangle ABC$. If the area of $\triangle ABC$ is 20 sq. units, then the area of $\triangle PQR$ is
 1) 100 sq. units 2) 5 sq. units 3) 10 sq. units 4) 3 sq. units
- The area of the triangle formed by $(a, b+c), (b, c+a)$ and $(c, a+b)$ is
 1) 0 sq. units 2) $\frac{a^3 + b^3 + c^3}{2}$ sq. units
 3) $\frac{a^2 + b^2 + c^2}{2}$ sq. units 4) $\frac{a+b+c}{2}$ sq. units

12. If $(1,2)$, $(3,4)$ and $(5,7)$ are the mid points of the sides of a triangle, then the area of the triangle is
 1) 4 sq.units 2) 1 sq.units 3) 2 sq.units 4) 3 sq.units
13. The equation of the line parallel to the line $ax + by + c = 0$, and passing through the point (a,b) is
 1) $ax + by = a^2 + b^2$ 2) $ax + by = 0$ 3) $bx + ay = a^2 + b^2$ 4) $bx + ay = 0$
14. The equation of the line perpendicular to the line $ax + by + c = 0$, and passing through the point (a,b) is
 1) $ax + by = a^2 + b^2$ 2) $bx - ay = 0$ 3) $bx + ay = 0$ 4) $bx - ay = a^2 - b^2$
15. The sum of the intercepts cut off by the following lines between the axes $x + y = a$, $x + y = ar$, $x + y = ar^2$,, where $a \neq 0$ and $r = \frac{1}{2}$ is
 1) $2a$ 2) $a\sqrt{2}$ 3) $2\sqrt{2}a$ 4) a
16. The equation of the line passing through $(-2,3)$, and having intercepts equal in magnitude but opposite in sign is
 1) $x - y - 5 = 0$ 2) $x - y + 5 = 0$ 3) $x + y - 1 = 0$ 4) $x - y + 7 = 0$
17. The equation of a line having intercepts a, b on the axes such that $a + b = 5$, $ab = 6$ is
 1) $2x + 3y - 6 = 0$ 2) $3x + 2y - 6 = 0$ 3) $2x + 10y - 5 = 0$ 4) $10x + 2y - 5 = 0$
18. The slope of the line passing through the points $(2, \sin\theta)$, $(1, \cos\theta)$ is 0, then general solution of θ is
 1) $n\pi + \frac{\pi}{4}, \forall n \in \mathbb{Z}$ 2) $n\pi - \frac{\pi}{4}, \forall n \in \mathbb{Z}$ 3) $n\pi \pm \frac{\pi}{4}, \forall n \in \mathbb{Z}$ 4) $n\pi, \forall n \in \mathbb{Z}$
19. The equation of a line passing through the points $(at_1^2, 2at_1)$, $(at_2^2, 2at_2)$ is
 1) $y(t_1 + t_2) = 2x - 2at_1t_2$ 2) $x(t_1 + t_2) = 2y + 2at_1t_2$
 3) $x(t_1 + t_2) = 2y - 2at_1t_2$ 4) $y(t_1 + t_2) = 2x + 2at_1t_2$
20. Find the equation of the straight line passing through the point of intersection of the lines $x + y = 5$, $x - y = 1$ and having slope -4, and also find the area of the triangle formed by this line with the coordinate axes.
21. Find the equation of the straight line perpendicular to the line $3x + 4y = 5$ and crossing Y-axis at $(0, -3)$, and also find the area of the triangle formed by this line with the coordinate axes.
22. Find the equation of the straight line, whose intercepts are a, b so that $a + b = 4$, $a - b = 2$ and also find the area of the triangle formed by this line with the coordinate axes.
